MASTERKIDS

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 37

Level 2 — Intermediate · Geometry II

Class	**Class 4 (Primary · Stage 2)**	Subject	**Mathematics**
Level	**Level 2 — Intermediate**	Topic Focus	**Geometry II — Perimeter & Area Applications**
Questions	**20**	Time Allowed	**35 minutes**
Total Marks	**20**	Worksheet No.	**37 of 100**
Curriculum	**CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned**		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1–5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 37 is intermediate perimeter-and-area at its best: wire reshaped from square to rectangle, same-perimeter different-area comparisons, joined squares (where edges vanish), photo-on-sheet leftover area, the inside border puzzle, and the 10,000-tiny-squares surprise. These are precisely the Achievers-style setups that appear in real IMO papers.

Learning Objectives

- Use perimeter and area formulas forwards and backwards.
- Track perimeter when wire is reshaped or shapes are joined.
- Compare areas of shapes with equal perimeters.
- Subtract areas (photo on sheet, patch in garden, inner border).
- Apply area to tiling cost and fencing rounds.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

SECTION A · Multiple Choice Questions · 20 Questions

Q1. A rectangle is 15 cm long and 8 cm wide. Its perimeter is:

(A) 23 cm
(B) 46 cm
(C) 120 cm
(D) 30 cm

Q2. The same rectangle (15 cm × 8 cm) has an AREA of:

(A) 46 sq cm
(B) 120 sq cm
(C) 23 sq cm
(D) 115 sq cm

Q3. A wire forming a square of side 9 cm is straightened and bent into a rectangle of length 10 cm. What is the rectangle's WIDTH?

(A) 6 cm
(B) 7 cm
(C) 8 cm
(D) 9 cm

Q4. A square and a rectangle have the SAME perimeter of 24 cm. The square's side is 6 cm; the rectangle is 8 cm long. Which has the BIGGER area?

(A) The square (36 vs 32 sq cm)
(B) The rectangle (32 vs 36 sq cm)
(C) Both equal
(D) Cannot compare

Q5. Find the perimeter of this rectangle.

(A) 20 m
(B) 40 m
(C) 84 m
(D) 48 m

Q6. A square field has an area of **49 sq m**. Each side is:

(A) 6 m
(B) 7 m
(C) 8 m
(D) 12 m

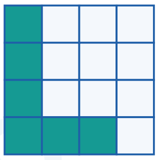
Q7. A rectangular hall 12 m long and 9 m wide is tiled with 1 m × 1 m tiles costing ₹50 each. The TOTAL tile cost is:

(A) ₹5,400
(B) ₹4,500
(C) ₹5,000
(D) ₹6,400

Q8. A farmer fences a square field of side 35 m with 3 ROUNDS of wire. How much wire does he need?

(A) 140 m
(B) 280 m
(C) 420 m
(D) 350 m

Q9. Each small square is 1 sq cm. What is the area of the shaded L-shape?



(A) 5 sq cm
(B) 6 sq cm
(C) 7 sq cm
(D) 8 sq cm

Q10. The perimeter of a square is 64 cm. Its AREA is:

(A) 16 sq cm
(B) 64 sq cm
(C) 256 sq cm
(D) 128 sq cm

Q11. A rectangle has area 72 sq cm and width 6 cm. Its LENGTH is:

(A) 12 cm
(B) 66 cm
(C) 36 cm
(D) 9 cm

Q12. The same rectangle (length 12 cm, width 6 cm) has a PERIMETER of:

(A) 18 cm
(B) 36 cm
(C) 72 cm
(D) 30 cm

Q13. Two identical squares of side 5 cm are joined edge to edge to form a rectangle. The PERIMETER of the rectangle is:

(A) 40 cm
(B) 30 cm
(C) 20 cm
(D) 25 cm

Q14. A photo 20 cm × 15 cm is stuck on a sheet 30 cm × 25 cm. What AREA of the sheet remains UNCOVERED?

(A) 300 sq cm
(B) 450 sq cm
(C) 750 sq cm
(D) 150 sq cm

Q15. A rectangular park is 50 m long and 30 m wide. Mira jogs around it TWICE. How far does she jog?

(A) 160 m
(B) 320 m
(C) 240 m
(D) 300 m

Q16. Which has the GREATER perimeter: a square of side 8 cm or a rectangle 10 cm × 5 cm?

(A) The square (32 vs 30 cm)
(B) The rectangle (30 vs 32 cm)
(C) Both equal
(D) Cannot say

Q17. A square cabbage patch of side 4 m sits inside a square garden of side 10 m. What area of the garden is NOT cabbage?

(A) 84 sq m
(B) 96 sq m
(C) 100 sq m
(D) 16 sq m

Q18. The length of a rectangle is DOUBLE its width. The width is 7 cm. The PERIMETER is:

(A) 28 cm
(B) 42 cm
(C) 21 cm
(D) 98 cm

Q19. A 1 m square is cut into 1 cm squares. How many tiny squares are made? (1 m = 100 cm)

(A) 100
(B) 1,000
(C) 10,000
(D) 100,000

Q20. A border of width 1 m runs around the INSIDE of a 12 m × 10 m garden. The inner lawn (without the border) measures:

(A) 11 m × 9 m
(B) 10 m × 8 m
(C) 12 m × 10 m
(D) 10 m × 9 m

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	C	A	B	B	A	C	B	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	B	B	B	B	A	A	B	C	B

Step-by-Step Solutions

Q1. Correct answer: (B)
2 × (15 + 8) = 2 × 23 = **46 cm**. (120 is the area.)

Q2. Correct answer: (B)
15 × 8 = **120 sq cm**.

Q3. Correct answer: (C)
Wire length = square perimeter = 4 × 9 = 36 cm. Rectangle: 2 × (10 + w) = 36 → 10 + w = 18 → w = **8 cm**.

Q4. Correct answer: (A)
Square area = 6 × 6 = 36. Rectangle: width = 24÷2 − 8 = 4, area = 8 × 4 = 32. **The square wins (36 > 32)** — for the same perimeter, a square always encloses more area.

Q5. Correct answer: (B)
2 × (14 + 6) = 2 × 20 = **40 m**.

Q6. Correct answer: (B)
Side × side = 49, and 7 × 7 = 49 → side = **7 m**.

Q7. Correct answer: (A)
Tiles needed = area = 12 × 9 = 108. Cost = 108 × 50 = **₹5,400**.

Q8. Correct answer: (C)
One round = 4 × 35 = 140 m. Three rounds = 3 × 140 = **420 m**.

Q9. Correct answer: (B)
Count the shaded squares: 4 down the side + 2 along the bottom = **6 sq cm**.

Q10. Correct answer: (C)
Side = 64 ÷ 4 = 16 cm. Area = 16 × 16 = **256 sq cm**.

Q11. Correct answer: (A)
Length = area ÷ width = 72 ÷ 6 = **12 cm**.

Q12. Correct answer: (B)
2 × (12 + 6) = **36 cm**.

Q13. Correct answer: (B)
The rectangle is 10 cm × 5 cm. Perimeter = 2 × (10 + 5) = **30 cm**. (NOT 40 — the joined edges disappear inside!)

Q14. Correct answer: (B)
Sheet = 30 × 25 = 750; photo = 20 × 15 = 300. Uncovered = 750 − 300 = **450 sq cm**.

Q15. Correct answer: (B)
Perimeter = 2 × (50 + 30) = 160 m. Twice = **320 m**.

Q16. Correct answer: (A)
Square: 4 × 8 = 32 cm. Rectangle: 2 × (10 + 5) = 30 cm. **The square (32 > 30)**.

Q17. Correct answer: (A)
Garden = 100 sq m; patch = 16 sq m. Rest = 100 − 16 = **84 sq m**.

Q18. Correct answer: (B)
Length = 14 cm. Perimeter = 2 × (14 + 7) = 2 × 21 = **42 cm**.

Q19. Correct answer: (C)
100 rows × 100 columns = **10,000** small squares.

Q20. Correct answer: (B)
The border takes 1 m off EACH side: length 12 − 2 = 10 m, width 10 − 2 = 8 m → **10 m × 8 m**.